

ADVANCED: ROUTINE CONSTRUCTION AND PROGRESSION

ADVANCED: ROUTINE CONSTRUCTION

Moving into the *advanced range* brings with it a new set of issues:

- Your training will become even more geared toward your specific sport or training.
- Shoring up weak links becomes vastly more important in order to progress in training. Many people have a weak link in their backs or even their quads if they are posterior chain dominant. Specific isolation work may be effective. If, for example, your back tends to be vastly stronger than your arms, biceps curls or other biceps exercise may be useful to shore up the weak links and improve your strength overall. The holds true for barbell lifts like the deadlift, which uses the legs, hips, and back extensively.
- While sleep quality, nutrition, and eliminating stress are important at every level, being on top of these health issues will help immensely at the advanced stage. Even 1% improvement will add up when you discover that increasing your strength and/or muscle mass starts to become difficult.
- Understanding how your body responds to training is important. At this stage, having a training log comes in handy, as it does at every stage of training. You can look back over how your body responded to certain rests, deloads, and intensity/volume of exercises. This makes the planning of workouts over the course of several weeks much easier.

Programming for the advanced level centers on a transition to the impressive B-skills and the need for more complex programming. You will learn to fully implement DUP protocol and other push/pull modifications to routines, as this type of training is the most effective at this point (though the traditional programming can be still used).

Here follow some examples of DUP protocol actually used in training. The first program was successfully implemented in 2006. The other exercises show examples of how to implement the DUP protocol on a similar repetition but variable weight scheme. (In this case, a makeshift weight belt was designed using a piece of rope with weights hooked onto it to provide additional resistance; an alternative would be a weighted vest.)

Daily Undulated Periodization (DUP) routine performed 3x per week. (Note: this athlete did not have access to a gym every day, so the days were not as consistent as they would have been with a traditional M/W/F program.)

The goal was iron cross, so iron cross pulls were the primary focus.

Monday

- 3x10 *Cross Pulls*
- 3x10 *Archer Rows*
- 3x10 *Straddle FL Rows*
- 3x10 *Hanging Leg Lifts*
- 3x10 *Pistols*

Wednesday

- 3x7 *Cross Pulls* + 15 lbs.
- 3x7 *Archer Rows* + 15 lbs.
- 3x7 *Straddle FL Rows* + 15 lbs.
- 3x7 *Hanging Leg Lifts* + 6 lbs.
- Deadlifts for 1x7 + 110 lbs., 1x7 + 170 lbs., and 1x7 + 200 lbs.

Saturday

- 3x4 *Theraband Cross Pulls*
- 3x4 *Straddle FL Rows* + 25 lbs.
- 3x4 *Archer Rows* + 25 lbs.
- 3x4 *Hanging Leg Lifts* + 10 lbs.
- 3x4 *Pistols* + 15 lbs.

Monday

- 3x10 *Cross Pulls* + 10 lbs.
- 3x10 *Straddle FL Rows* + 15 lbs.
- 3x10 *Archer Rows* + 15 lbs.
- 3x10 *Pistols* + 15 lbs.

Thursday

- 3x7 *Cross Pulls* + 20 lbs.
- 3x7 *Straddle FL Rows* + 23 lbs.
- 3x7 *Pistols* + 20 lbs.
- + Conditioning

Friday

- 3x4 *Cross Pulls* + 30 lbs.
- 3x4 *Straddle FL Rows* + 40 lbs.
- 3x4 *Archer Rows* + 40 lbs.

- 3x4 *Ring Dips* + 40 lbs.
- *Pistols* for 1x4 + 30 lbs and 2x4 + 40 lbs.

Tuesday

- 3x10 *Cross Pulls* + 20 lbs.
- 3x10 *Ring Dips* + 40 lbs.
- 3x10 *Ring Dips* + 40 lbs.
- 1x10 *Straddle FL Rows* + 30 lbs.
- + Conditioning

Note the increase in weight compared to the previous 10 RM, 7 RM, and 4 RM workouts. Attempt was made to keep exercises consistent with the repetitions but modify the weight accordingly (a good level of success was achieved). This allowed for large increases in weight on each exercise the following week: cross pulls, archer rows, and straddle FL rows all increased by 10-15 pounds after a week at 3x10 (from the first Monday to the second Monday). This is a substantial strength gain.

This particular athlete had poor nutrition at this time so the progress devolved after the first two weeks (which is why the entire cycle is not shown). Ultimately, cross pulls with an additional 50 pounds were achieved (in another cycle a few months later using a similar method.)

Here is a ~2 day per week program of light/heavy workouts. (Iron cross is again the ultimate goal.)

- Tuesday – *Cross Pulls* +10 lbs: 3x10
- Friday – *Cross Pulls* +15 lbs: 3x5
- Saturday – Miscellaneous Workout
- Tuesday – *Cross Pulls* +15 lbs: 3x10
- Friday – *Cross Pulls* +25 lbs: 3x5
- Tuesday – *Cross Pulls* +20 lbs: 3x8
- Friday – *Cross Pulls*: 1x8 bodyweight, 4x4 +35 lbs.
- Tuesday – *Cross Pulls*: 1x8 bodyweight, 3x8 +25 lbs.
- Thursday – *Cable Cross Pulls*: 4x5 with a setting of 6
- Tuesday – *Cross Pulls*: 1x8 bodyweight, 1x5 20 lbs, 3x4 40 lbs.
- Friday – *Cross Pulls*: 1x8 bodyweight, 1x4 25 lbs, 2x3 47.5 lbs., 1x3 50 lbs.
- Tuesday – *Cross Pulls*: 1x8 bodyweight, 1x4 25 lbs, 3x3 50 lbs.

Notice the manipulation of volume, alternating from 10 RM and 5 RM. As the cycle progresses, you build toward 8 RM and 4/5 RM, which biases the workout more toward strength. By the end, you are performing a 3 RM workout. This is a modification you can make with light/heavy or DUP, which is similar to classic periodization with its hypertrophy, strength, and power cycles. Instead of sticking with 10 RM, 7 RM, and 4 RM, you can instead run a hypertrophy scheme at first, slowly biasing toward strength at the end. This is extremely beneficial because the modern DUP protocol is better than traditional periodization at maintaining attributes. This program was implemented to focus on strength protocol in order to perform the iron cross for a swiftly-upcoming event.

This is another actual program, this one implemented with weighted dips. It was not the only pushing exercise in the program, but it gives an example of how a single exercise can be structured to make progress.

Notations are in *weight x reps x sets*. If there are only two numbers, it is just *weight x reps*. An 'F' placed after a repetition denotes that it was a failed repetition.

- 5/27 – BWx10, 60x5, 80x5x4
- 5/28 – BWx10, 60x5, 80x5, 100x5x3
- 5/30 – BWx10, 60x5, 80x3, 120x4x3
- 5/31 – BWx10, 20x3, 80x3, 120x3, 130x3x3
- 6/2 – BWx10, 60x5, 80x3, 120x3, 140x3x3
- 6/3 – BWx5, 60x3, 80x3, 120x4+1 (rest-pause), 130x4x3
- 6/4 – BWx5, 60x3, 120x3, 180x1x7 (eccentrics)
- 6/7 – BWx10, 60x5, 120x3, 130x5x3
- 6/9 – BWx10, 70x3, 130x3, 150x3x2, 160x2 (PR)
- 6/11 – BWx5, 55x3, 110x3, 140x2, 160x1F, 160x1, 165x1F
- 6/14 – BWx5, 60x5, 120x3, 150x3x3
- 6/16 – BWx5, 70x3, 90x3, 130x2, 150x1, 165x1F, 165x1 (PR), 170x1 (PR), 175x1F

This was all performed at a bodyweight of 135 pounds. Thus, 170 pounds were added to 135 pounds of the athlete. The previous 1 RM was +155 pounds, and weighted dips had not been performed for a couple of months. During a three-week ramp-up with the above program, 15 pounds were added to this weighted dip. This proves how effective this protocol can be. Not many athletes can add such a significant amount of weight at such an advanced level of strength. (This does not even take into account the brevity of the ramp-up period.)

Note the consistent scheme of decreasing the repetitions and increasing the weight with almost every workout. The following RM progressions were followed on a day-to-day basis: 5, 5, 4, 3, 3, 4, heavy eccentrics (similar to a 1-2 RM), 5, 2, 1, 3, 1. Lowering weight and adding repetitions serves to add volume, which helps to force adaptations. You can then aim to max and hit a PR, even if you are advanced in strength.

By the way, this program was by no means optimal. Ramping up toward a 1 RM this quickly and in this way is incredibly taxing and does not provide as much volume as a strength stimulus. A better progression would be to start out with more repetitions and stick with 2-3 RM. Start out with 8 RM, then move to 6 RM, 4 RM, and slowly ramp down to 2-3 RM. However, as flawed as this repetition scheme was, it clearly demonstrates the potential of a hybridized DUP system at an advanced level of strength.

Another example is a push/pull system integrated with light/heavy work. This is an advanced system that focuses on week-to-week gains in each exercise while separating push and pull to add additional volume. This system can be particularly effective for hypertrophy because of the additional volume. A push/pull routine performed 4-5x per week may look like the typical M/Tue/Thur/F with push being on M/Thur and pull being on Tue/F. Here follows an example:

- Monday: Heavy Push
- Tuesday: Light Pull
- Wednesday: Rest
- Thursday: Light Push
- Friday: Heavy Pull
- Saturday: Rest
- Sunday: Rest

The exercises on the heavy days consist of 3-5 repetitions, while the exercises on the light days consist of 5-8 repetitions. You can bias this routine toward strength by lowering the repetition ranges. This would make the exercises on your heavy and light days 1-5 and 3-6 repetitions, respectively. If hypertrophy is your goal, simply skew the numbers of repetitions in the opposite direction. This would make the exercises on your heavy and light days 3-8 and 6-12 repetitions, respectively. Alternating your focus each week is completely acceptable and even encouraged if you hit a plateau.

Heavy Push

- 3x4 *Straddle Planche Pushups*
- 3x5 *Rings Freestanding HSPUs*
- 4x5 *90-degree Rings-Turned-Out (RTO) Dips*
- + Legs

Light Pull

- 3x8-10 *Straddle Front Lever Pull-ups*
- 3x8 *Pulley-assisted (20-lb. assist) OAC (or Iron Cross variations)*
- 3x6 *German Hang Pullouts*
- + Legs

Light Push

- 3x10 *Advanced Tuck Planche Pushups*
- 2x7 *Rings Freestanding HSPUs*
- 3x7 *90-degree Rings-Turned-Out (RTO) Dips*
- + Legs

Heavy Pull

- 3x3 *Full Front Lever Pull-ups*
- 3x4 *Pulley-assisted (10-lb. assist) OAC (or Iron Cross variations)*
- 3x3 *German Hang Pullouts* + 1 lb. ankle weights
- + Legs

This routine was structured using many different light/heavy variations. With some exercises, like the planche, it is very difficult to add weight effectively. There are multiple ways to work around this; the easiest is to use some form of added resistance in the form of weight belt, weighted vest, or other makeshift item. Alternatively, the subsequent progression may be made easier by using bands. Even without added weight the progressions themselves work well to base your light/heavy, DUP, or another workout structure on.

Say you are stuck at a transition where you can hold a straddle planche for five to six seconds during a “heavy” day, but cannot work higher times with the straddle planche to make it effective for a “light” day. You can always move down to the previous progression for the lighter day (the advanced tuck planche in this instance). Train this exercise with twenty-second holds on your light day. If you are training toward the planche on the floor or parallettes, it can be beneficial to move to the rings for your light day when you move down a progression. You could even use this as alternative supplementary work.

As you can see from the programming illustrated above, multiple factors are used to distinguish a light day from a heavy day. They are based on what equipment may or may not be available. Be creative!

- To change a heavy day into a light day for rings HSPUs and RTO dips, decrease the number of sets and increase the number of repetitions per set.
- To go from a heavy day to light day with planche pushups, switch from the higher progression (straddle planche pushup) to the lower progression (advanced tuck planche pushup). You can do the same for front lever pull-ups.
- For German hang pullouts and one-arm chin-up work, you can use weights to make the heavy day more difficult; just decrease the number of repetitions per set.

If you are on a DUP protocol with light, medium, and heavy days you can use these same steps to adjust those days in order to progress effectively. When you progress to the upper portion of the advanced range, your gains tend to appear once per week with DUP but may begin to taper off to once every two weeks. Do not be upset if it takes an entire cycle to see even a small improvement in strength. An increase in complexity of programming will be needed if you want to continually progress at this level. At this point, training becomes a bit more instinctive, as you have developed a ridiculous amount of strength and overall knowledge of yourself and what works best for you.

The term auto-regulation and the style of training implied is how you will train from here on out. As discussed in the programming chapters, you want to have quality workouts. You know how your body handles stress and you will know if you can do more or need to stop on any given day. At this level, it is necessary to add or subtract exercises or sets mid-workout based on how your body feels. The best way to track a workout and its effect on your body is using the RPE system discussed in the chapter on overreaching and overtraining. It becomes very important to track how you are feeling in workouts, as your progress will not be as obvious from workout to workout. Do not try to rely on your memory alone. Take good notes. They will be beneficial for years to come.

At this level of training, you can finally try a light/heavy or DUP-type taper. Start with a repetition scheme of 12/6 for light/heavy, respectively. Every week, change the number of repetitions in your light and heavy workouts until you “taper” to hit a 1 RM at the end of the cycle. Week 1 at 12/6 RM, Week 2 at 10/5 RM, Week 3 at 8/4 RM, Week 4 at 6/3 RM, and Week 5 finishes with a 4/1-2 RM to set a *personal record* (PR). Sets should be held constant at three sets or increased as you move toward the heavier days, if you can handle the volume.

A hybridized DUP program on a push/pull system performed 4x per week may look like this:

Week 1

Heavy Push

- 3x6 *Straddle Planche Pushups*
- 4x6 *Rings Freestanding HSPUs*
- 3x6 *90-degree RTO Leaned-Forward Dips*
- + Legs

Light Pull

- 3x10 *Straddle Front Lever Pull-ups*
- 3x10 *Pulley-assisted* (20 lb. assist) *OAC* (or *Iron Cross* variations)
- 3x10 *German Hang Pullouts*
- + Legs

Light Push

- 3x10 *Rings Advanced Tuck Planche Pushups*
- 3x10 *Rings Freestanding HSPUs*
- 3x10 *90-degree RTO Dips*
- + Legs

Heavy Pull

- 3x6 *Full Front Lever Pull-ups*
- 3x6 *Pulley-assisted* (5-lbs assist) *OAC* (or *Iron Cross* variations)
- 3x6 *German Hang Pullouts*
- + Legs

Week 2**Heavy Push**

- 3x4 *Straddle Planche Pushups* (+ 5 lb. vest or ankle weights)
- 4x4 *Rings Freestanding HSPUs* (+ 5 lb. vest or ankle weights)
- 3x4 *90-degree RTO Leaned-Forward Dips* (or change progression to e.g. *Dip Straight-Body Press to Handstand*)
- + Legs

Light Pull

- 3x7 *Full Front Lever Pull-ups*
- 3x7 *Pulley-assisted* (5-lbs assist) *OAC* (or *Iron Cross* variations)
- 3x7 *German Hang Pullouts*
- + Legs

Light Push

- 3x7 *Rings Advanced Tuck Planche Pushups*
- 3x7 *Rings Freestanding HSPUs*
- 3x7 *90-degree RTO Dips*
- + Legs

Heavy Pull

- 3x4 *Full Front Lever Pull-ups* (with weighted vest)
- 3x4 *Unassisted OAC* (or *Iron Cross* variations)
- 3x4 *German Hang Pullouts* (with ankle weights)
- + Legs

In Week 2, the light workouts are very similar to the heavy workouts from the previous week. You can make them slightly harder or easier, depending on how well you are progressing from week to week. The main objective is to progress.

Week 3

Heavy Push

- 5x3 *Straddle Planche Pushups* (+ 10 lb. vest or ankle weights)
- 5x3 *Rings Freestanding HSPUs* (+ 10 lb. vest or ankle weights)
- 3x3 *Dip Straight-Body Press to Handstand*
- + Legs

Light Pull

- 3x5 *Full Front Lever Pull-ups* (with weighted vest)
- 3x5 *Unassisted OAC* (or *Iron Cross* variations)
- 3x5 *German Hang Pullouts* (with ankle weights)
- + Legs

Light Push

- 3x5 *Straddle Planche Pushups* (+ 5 lb. vest or ankle weights)
- 3x5 *Rings Freestanding HSPUs* (+ 5 lb. vest or ankle weights)
- 3x5 *90-degree RTO Leaned-Forward Dips* (or change progression to e. g. *Dip Straight-Body Press to Handstand*)
- + Legs

Heavy Pull

- 3x3 *Full Front Lever Pull-ups* (+ 10 lb. weighted vest)
- 3x3 *OAC* (+ 5 lbs.)
- 3x3 *German Hang Pullouts* (with ankle weights)
- + Legs

Aim to see the same progress from Week 2 to Week 3 that you saw from Week 1 to Week 2. This is an example of a routine where you can progress from week to week, even as an advanced athlete.

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Modifying Routines: Sometimes workouts do not go as planned and progress is slower or faster than you expected. This can be affected by many factors aside from your actual workouts. It is difficult to know if programs will work at higher levels until you have significant experience. However, since you have come this far, you are likely experienced enough to have a “feel” for how a routine will progress. If you need to modify the number of repetitions to make your workout more or less difficult, you can trust yourself to make this call. Just be sure to keep your light days light and your heavy days heavy. That is the most important part of a split light/heavy routine.

Keep in mind that as the stimulus to force adaptations increases and your training increases to match it, it is very easy to start doing too much rather than too little. This situation is partially addressed by properly scheduling deload weeks, but be aware that you may be accumulating unaddressed fatigue in your many years of training, even if you are still progressing.

Overreaching and Overtraining: After many years of consistent training, take an extended deload period of a few weeks to a few months. During this time, it is not necessary to fully rest the entire time. Just relax! Enjoy some time off to spend with your family, do something fun, or perhaps learn a new sport. If you have been training consistently for years, there will be very little overall decrease in gains when you come back. Instead, your overall total fatigue will have dissipated, leaving you fresh and ready to continue with your training.

One notable athlete had trained consistently for nearly four years with appropriate rest breaks, and his abilities were around Level 11-12 in most categories. He took a two-month extended break, during which time his body continued to experience changes. His hypertrophy increased as he “recovered” more. He had not been aware that his body even needed this extended break, as he had been making good progress beforehand.

Sometimes when our abilities skyrocket, we lose sight of the other qualities that are important to training effectively. Life is not just a game to see who can be the best. We should really have a love for what we are doing, and this sometimes means taking a break. Rest can help you regain your appreciation for training, which can often be lost in the day-to-day grind. This fresh perspective often positions you for success when you return.

Utilizing Periodization: Do not be afraid of failure. Even if you do fail, at least you tried! Many of the periodization and programming schemes that you will try out in your early and intermediate, and even into the beginnings of the advanced stages will be mediocre. However, that experience will help you learn how your body responds to various exercises and periodization methods. It was only as you gain more experience that you will be able to analyze your programming and refine it to be more effective. You will learn that it is important to put forth the effort to experiment with your training to learn what will work well for you, especially if your progress has stalled.

If you are a coach, this will pay dividends for your athletes. Be flexible, as your athletes are not you and you are not your athletes. There is a general, foundational level of knowledge that coaches must learn in order to know what works for their sport. This fundamental knowledge can be used effectively for most athletes. However, the mark of a great coach is the ability to build upon these fundamentals and tailor an approach for each individual athlete based on their unique response to the training program. Unlike cookie-cutter programs, the goal of *Overcoming Gravity* is to teach you to identify what works best for you so you can build your own routine.

Goals: At the advanced training level, your goals need to become even more specific. Maintain the strength level of the goals you have already achieved while simultaneously working toward one push or one pull goal at a time. This enables you to focus *all* of your push and pull exercises toward your one specific goal. Such focus on one particular goal will help to drastically increase the rate at which you progress.

Consider Outside Factors: It is also very important to be aware of your sleep quality, nutrition, and stress levels. You will not get very far without having these factors in order. The easiest example to use is that of

David Brailsford and the British cycling team, who only won 2 gold medals from 1908-2002 but won 58 gold medals in world championships from 2003-2013 and 8 gold medals in the 2008/2012 Olympics. Basically, Coach Brailsford operated on the *marginal gains theory* in which he aimed to improve his athletes in every aspect in 1% increments.

If you can improve your workouts, sleep quality, nutrition, stress levels, and other factors by just 1%, these slight improvements will eventually accrue large improvements. It may seem like a negligible improvement to take off 0.01 seconds by improving performance through various factors here and there, but when you make 10 of these improvements it equals 0.10 seconds. In swimming, track and field, and cycling, 0.10 seconds can be the difference between first and fourth place.

Training Log: Keep one, and record RPE values too. Track changes in daily performance and how you felt while training. A written log will allow you to see if you are undertraining or overtraining, which is critical at this level. As an advanced athlete, the window for effective volume, frequency, and intensity narrows. Use information from your training log to modify your routines as necessary.

Weak Links: The advanced progressions are where weak links become much more pronounced. Check every few weeks to see if you need extra work in particular areas. Rely on your training log. The extremely difficult movements in an advanced workout place a large amount of strain on the areas being worked. Your weak links may be identified for you by the exercises you are performing. For example, you know a one-arm chin-up or iron cross is working your back muscles. However, if you are not progressing, you may need to focus your work in one of two places. Your back muscles may not be taxed enough due to a lack of volume. Alternatively, your biceps or other areas may be the weak link that is preventing progression. You may need to add a bit of isolation work to bring up these weak links. Experiment, take notes, and discover what will help you progress.

Overuse Modifications: When you are dealing with very difficult bodyweight exercises your body will usually tell you if an exercise is too much by the strain on your muscles or connective tissues. Listen closely to your body because higher intensity exercises have the tendency to injure you more, especially when you are fatigued. If you feel strain on an area, you should definitely not go to failure. As a reminder, here is the modification information from previous chapters:

- Eliminate, reduce volume, or substitute exercises.
- Add additional prehabilitation, isolation, mobility, or flexibility work as needed for the particular area.

Avoiding Injuries: It pays to remember that *the number one predictor of an injury is a previous injury*. Always back down and modify your routine based on how your body responds—especially in the context of potential overuse injuries. An injured athlete always progresses slower than one who stopped short of injury!

And Beyond: Beyond Level 12, athletes have diverse goals they want to achieve and the experience to make it happen. While this book covers a lot of ground, not much time is spent discussing special techniques to make workouts shorter, such as pairing exercises or various types of split routines. If you have further questions in that area, the Internet is a great resource. Your questions may even help another athlete! If you are at a complete loss or just want to continue the discussion, post on this Reddit forum: www.reddit.com/r/overcominggravity.